Which cell is a type of neuron?

A. Basal cells

B. Brush cells

C. Olfactory cells

D. Sustentacular cells

E. Alveolar type I cell

C

What type of epithelium lines the trachea?

A. Simple squamous epithelium

B. Simple cuboidal epithelium

C. Simple columnar epithelium

D. Stratified squamous epithelium

E. Psudo-stratified columnar epithelium

E

A unique feature of the olfactory epithelium is the presence of ?

A. Goblet cells

B. Neurons that are replaced throughout life

C. Hair cells

D. Receptors for taste

E. Clara cells

B

Which cell is a respiratory macrophage?

A. Kupffer cells

B. Histiocyte

C. Dust cell

D. Langerhans cell

E. Microglia

What is the smallest airspace of the respiratory tree?

A. Alveolar duct

B. Respiratory bronchiole

C. Alveoli

D. Terminal bronchiole

E. Alveolar sac

Example #1

pH	7.50
pCO2	29
HCO3	22

Respiratory Alkalosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Example #2

pH	7.25
pCO2	60
HCO3	26

Respiratory Acidosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Example #3

pH	7.34
pCO2	60
HCO3	31

Respiratory Acidosis with Metabolic Compensation

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Example #4

pH	7.50
pCO2	48
HCO3	36

Metabolic Alkalosis with partial compansation

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Example #5

pH	7.20
pCO2	21
HCO3	8

Metabolic Acidosis with Respiratory partial Compensation

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Example #6

pH	7.35
pCO2	49
HCO3	30

Respiratory acidosis compelete Compensation

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Example #7

pH	7.45
pCO2	48
HCO3	35

Metabolic alkalosis compelete Compensation

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	□7.35-7.45□	Alkalemia
pCO2	Respiratory alkalosis	□35 - 45□	Respiratory acidosis
HCO3	Metabolic acidosis	□22 – 26□	Metabolic Alkalosis

Case 1

A 55-year-old patient with a history of COPD "Chronic obstructive pulmonary disease" presented to the hospital with shortness of breath, confusion, drowsiness, leathargy.

On admission, she was found to have the following



Laboratory investigations:

-Oxygen saturation(SaO2)= 90%

-Arterial blood gas: PH = 7.31

Pco2= 85 mm Hg

Bicarbonate= 24 mEq/L

What is your provisional diagnosis ?

Respiratory acidosis



MCQ

- **1. Choose the etiology of atypical pneumonia:**
- **A. *Haemophilus influenzae***
- **B. *Mycobacterium tuberculosis***
- **C. *Mycoplasma pneumoniae***
- **D. *Streptococcus pneumoniae***
- **E. *Pseudomonas aeruginosa***

- **2. 65 years old alcoholic man had sudden fever, chills, pleuritic pain & production of rusty sputum, he has infection caused by:**
- **A. *Streptococcus pneumoniae***
- **B. *Haemophilus influenzae***
- **C. *Mycoplasma pneumoniae***
- **D. *Escherichia coli***

1. C

2. A

1. E
2. D
3. C

- **How do you get TB?**
- A. Through the air
- B. Through sexual contact
- C. Through contaminated food
- D. Through blood
- E. A and C

- **What are the symptoms of active TB?**
- A. Weight loss
- B. Night sweats
- C. Loss of appetite
- D. All of the above

- **Within the body, TB organisms**
  A) are engulfed and killed by macrophages.
- B) reside indefinitely in an active stage in the body.
- C) are walled off by macrophages in encapsulated granulomas called tubercles.
- D) replicate and gain access to lymphatic channels through the blood stream.
-

Answer the following questions:

1-First line antituberculous drugs include all of the following except:

a-Ethionamide b-Isoniazide

c-Rifampin d-Pyrazinamide

2-Isoniazide has the following adverse effects:

a-Cardiotoxicity

b-Hepatotoxicity, peripheral neuropathy

c-Loss of hair

d-Immunotoxicity

1. A
2. B

MCQ

1. Fungi that are more virulent and cause disease in healthy hosts are classified as?

A. superficial
B. cutaneous
C. subcutaneous
D. systemic
E. opportunistic

2. Which one of the following fungi causes allergic bronchopulmonary disease resulting in marked elevation of serum IgE?

a. *Aspergillus fumigates*
b. *candida albicans*
c. *Pneumocystis jiroveci*
d. *Histoplasma capsulatum*

3. Aspergillosis is best described as a:

A. Subcutaneous mycosis
B. Opportunistic mycosis
C. Dimorphic systemic mycosis
D. Cutaneous mycosis
E. Superficial mycosis

1. D
2. A
3. B

The immunoglobulin class most frequently responsible for
inhibition of bacteria on mucosal surfaces is:
(A) IgG
(B) IgM
(C) IgA
(D) IgE

Which one of the following cell types expresses receptors for
IgE on its cell surface that stimulate the cell to mount a response to an allergen?
(A) T cells
(B) B cells
(C) NK cells
(D) Mast cells

IgE-mediated histamine release is classified as what type of
hypersensitivity reaction?
(A) Type I
(B) Type II
(C) Type III
(D) Type IV

1. C
2. D
3. A

Questions dr hoda

1- in a healthy individual breathing spontaneously which of the following pressure is positive with reference to atmospheric pressure?

a. Alveolar pressure during inspiration

b. Alveolar pressure during expiration

c. Intrapleural pressure during inspiration

d. Intrapleural pressure during expiration

2- The maximum amount of gas that can be exhaled after full inspiration is called

a. Expiratory reserve volume

b. Vital capacity

c. Total lung capacity

d. Functional residual capacity

3- The largest volume of gas that can be moved into and out of the lung in 1 minute by voluntary effort called...

a. Respiratory minute volume

b. Minute ventilation

c. Maximum voluntary ventilation

d. Vital capacity

4- Intrapleural pressure is normaly negative because ...

Ans: chest wall and lung recoil in opposite direction at FRV

5- What is the effect of 2,3 BPG on affinity of Hg for oxygen?

a. Increase affenity

b. decrese affenity

6- which of the following has the highest p50?

a. Myoglobin

b. Hbf

c. Hba

7- All of the following shift oxyhemoglobin dissosiation curve to the right except

Ans : Co

8- decrease in o2 affenity of Hb called

Ans. Boher effect

9- Which of the following reduce synthesis of 2,3 BPG ?

a. Acidosis

b. At exercises

c. Anemia

d. In high attetiude

10- The amount of oxygen dissolved in plasma

Ans: Po2

Rapid Quiz:

- (1) Which of the following is (are) responsible for the high (GFR)?
- (A) High blood hydrostatic pressures in the glomerular capillaries
- (B) High permeability of the glomerular capillaries
- (C) High hydrostatic pressure of the fluid in the Bowman's capsule
- (D) The renin-angiotensin-aldosterone system
- (E) High rate of water reabsorption across the walls of the proximal tubules

- 2-Renin is produced and secreted from granules located in the:
  a- macula densa.
  b- mesangial cells.
  c- intercalated cells.
  d- afferent arteriole.
  e- interstitial cells.

- 4-Renal blood flow is kept constant over a wide range of arterial pressures (about 60 – 180 mmHg). What is this range of pressure referred to as?
  a- the myogenic pressure threshold.
  b- the autoregulatory range.
  c- the threshold range.
  d-extrinsic regulation range.
  e-neurogenic regulation range.

- 5- Which two factors are controlled by tubuloglomerular feedback?
  a- GFR and RBF.
  b- RBF and tubular load.
  c- GFR and tubular load.
  d- Filtration fraction and RBF.
  e- GFR and filtration fraction.

Upper Resp. Tract

Rhinoscleroma is a:

A. Benign epithelial tumor

B. Benign mesenchymal tumor

C. Chronic non-specific inflammation

D. Chronic specific inflammation

Rhinoscleroma is a:

A. Benign epithelial tumour

B. Benign mesenchymal tumour

C. Non-specific inflammatory lesion

D. Granulomatous lesion

Laryngeal nodule is:

A. Benign epithelial tumor

B. Benign mesenchymal tumor

C. Non-specific inflammatory lesion

D. Granulomatous lesion

The following lesion is precancerous for laryngeal carcinoma:

A. Rhinoscleroma

B. Laryngeal nodule

C. Squamous cell papilloma

D. Nasal polyp

The characteristic cells in rhinoscleroma are:

A. Mickulicz cells

B. Russel bodies

C. Langhan,s giant cell

D. Both a and b

Glottic carcinoma arising from true vocal cords is usually

A. Good prognosis

B. Moderately to poorly differentiated

C. Metastasize early

D. Presented late

The inflammatory cellular infiltrate in rhinoscleroma consists of:

A. Plasma cells

B. Russel bodies

C. Mickulicz cells

D. All of the above

The following lesion is precancerous lesions for laryngeal carcinoma:

A. Rhinoscleroma

B. Laryngeal nodule

C. Squamous cell papilloma

D. acute Rhinitis

The characteristic cells in rhinoscleroma are:

E. Mickulicz cells

F. Russel bodies

G. Langhan,s giant cell

H. Mickulicz cells and Russel bodies

I. Psommoma bodies

Glottic carcinoma arising from true vocal cords is usually

E. Well to moderately differentiated

F. poorly differentiated

G. Presented late

H. Of poor prognosis

The inflammatory cellular infiltrate in rhinoscleroma consists of macrophages with hydropic degeneration called:

E. Epitheloid cells

F. Russel bodies

G. Mickulicz cells

H. Langhans' giant cells

The inflammatory cellular infiltrate in rhinoscleroma consists of plasma cells with hyline degeneration called:

A. Epitheloid cells

B. Russel bodies

C. Mickulicz cells

D. Psommoma bodies

Bleeding from nose is called:

A. Epistaxis
B. hematemesis
C. Hemoptysis
D. Rhinorrhea

……. carcinoma arising in larynx has the best prognosis

A. Supraglottic
B. Infraglottic
C. Glottic
D. Supra and infra glottis

All are characters of atrophic chronic rhinitis except……

A. The mucosa is thickened and polypoid.
B. The nasal mucosa is covered by crusts of dried discharge which have offensive odour.
C. Atrophy of the nerve endings causing loss of smell (anosmia).
D. Atrophy of the nasal bones

Squamous Cell Papilloma in children ……..

A. Is usually solitary
B. caused by a viral infection.
C. Lacks tendency to recur after surgical removal,
D. may turn malignant.

All are true about Squamous Cell Papilloma in adult, except ……..

A. Is usually solitary

B. caused by a viral infection.

C. Lacks tendency to recur after surgical removal,

D. do not turn malignant

verrucous carcinoma is …….

A. poorly differentiated type of squamous cell carcinoma

B. This type is superficially invasive

C. usually metastasize

D. of unfavorable prognosis .

the supraglottic laryngeal carcinoma……

A. is the most common type of laryngeal carcinoma

B. Arises from false cord, ventricle and epiglottis.

C. The tumor is well to moderately differentiated.

D. Has good prognosis

the supraglottic laryngeal carcinoma……

A. Presented early

B. Well differentiated

C. Has slow rate of growth

D. Commonly associated with nodal metastasis.